

Theme 1

Sorting machine

Learning Outcomes

By the end of this theme, students will be able to:

1. Students will be able to differentiate between magnetic and non magnetic materials
 2. Know which material is the best to design a sorting machine
 3. Encourage creativity by designing a house made of selected materials.
 4. Using EDP to create a sorting machine
 5. Understand the concept of the pulley
-

UNIT 01: [Beginner]

Chapter 01: **Mechanics**

Lesson 01: **Sorting magnetic and non magnetic materials**

Time Frame: 3 hours

Lesson Objectives:

- .differentiate between materials
- .understand sorting recycle into different groups
- .introduction to soft and hard materials

Challenge questions of the lesson:

- .what was the school trying to be?
- .what would happen if we tried to build a house out of paper instead of bricks?
- .can you help me sort these materials into soft and hard ones?

ESL Relation:

Self awareness: understanding how to sort recycled materials.

Empathy: developing responsibility towards to planet to make it greener by recycling.

Social awareness: recognize the importance of recycling.

Answers of exercises

1. B, sort recycling items
2. C, excited
3. C, greener

Answers of exercises

Hard materials : wood, metal, brick, ice, plastic, glass, stone
Soft materials: wool, fabric, paper, leather, water

Lesson Plan

1. Engage

- **Teacher Role:**
facilitating discussions, asking questions, innovate competitions

- **Expected From Students:**
participating in discussions, sharing ideas

- **Parts of the books included.**

Lesson 1 (4 pages)

- **Intro Story:**

Use engaging strategies to grab students' attention. Story telling (adam went to a new mall which have high technology in sorting rubbish ,so his curiosity was aroused and started asking questions about difference in materials.

- **Story Reflection:**

- Questions

- What is the difference between materials?

- What is recycling?

- Can you test the suggested material if its magnetic or non-magnetic?

- Suggested answers

- hard, soft, heavy or light / paper, iron, rubber, plastic

- converting rubbish into usable tools to make the planet cleaner.

-use a magnet

- (List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)

-realia (materials and magnet)

2. Think

- **Questions and Possible Answers**

- **Brainstorm Web:**

[Provide space for ideas and connections.]

- (List strategies teachers can use to deliver this part, including the name of the strategy and how to implement it.)
-

3. Explore

- **Teacher Role:**

Let the students feel the texture of different materials

- **Expected From Students:**

Sorting different materials into 3 baskets.

Use magnet to identify magnetic materials

- **Parts of the books included.**

- **Activity - Do It:**

- **Materials Needed**

(pencil - plastic toy - rubber band - screw - paper - can - straw - copper - coin - magnet)

- **Step-by-Step Instructions**

- **Reflection Questions**

- Why are some materials attracted to the magnet and others not?

- can you describe the suggested material ?

- why do we have to sort the rubbish?

4. Explain

- **Teacher Role:**
[Explain concepts, guide discussions, etc.]
 - **Expected From Students:**
[Engage actively, share observations, debates]
 - **Parts of the books included.**
 - **Watch It:**
 - *Pre-Watching:* Set a purpose.
 - *During Watching:* Active engagement.
 - *Post-Watching:* Discussion and reflection.
 - **Read About It:**
 - *Pre-Reading:* Discussion.
 - *During Reading:* Engagement.
 - *Post-Reading:* Summarize and reflect.
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5. Elaborate

- **Teacher Role:**
[Encourage creativity and application of knowledge.]
- **Expected From Students:**
[Create models]
- **Parts of the books included.**
- **Assessment:**
- Focus (answer the questions and list some strategies to apply while answering them.)
- Practice. (waterproof materials game)
 - **Materials Needed**

(laptop - internet)

- **Step-by-Step Instructions**

Open this link (<https://www.sciencekids.co.nz/gamesactivities/materialproperties.html>)

- **Reflection Questions**

- Challenge: Left for the students to be made at home.

Can you make a model of a house using different materials?

6. Evaluate

- **Teacher Role:**
[Facilitate assessments and provide constructive feedback.]

- **Expected From Students:**

[Answer questions, demonstrate understanding, make projects.]

- **Parts of the books included.**

Showcase: Evaluate the students work in the main project and add a rubric so teachers can follow .

- ❖ **Relation to Standards:**

[Link to relevant educational standards.]

Theme 1

[Sorting magnetic and non-magnetic materials]

UNIT 01:

Chapter 01: [Mechanics]

- Lesson 02: Engineering Design Process
- Time frame: 15 mins.

Lesson Objectives:

By the end of this lesson, students will be able to:

1. **Define** the steps of the Engineering Design Process (EDP) and apply them to solve a real-world problem.
2. **Investigate** the properties of metals vs. non-metals to determine effective sorting methods.
3. **Design and prototype** a device (mechanical, magnetic, or sensor-based) to separate metals from mixed materials.
4. **Test and refine** their solution based on feedback and constraints.
5. **Collaborate effectively** in teams, demonstrating communication and problem-solving skill

Challenge Questions (To Guide Inquiry):

1. **Problem Identification:**
 - Why is it important to separate metals from other materials in recycling or manufacturing?
 - What are the key differences between metals and non-metals that we can use for sorting?
2. **Research & Brainstorming:**
 - What existing technologies (magnets, sensors, density separation) are used to sort metals?
 - How can we adapt these methods into a simple, cost-effective design?
3. **Design & Prototyping:**
 - What materials and mechanisms will you use in your design? (e.g., electromagnets, conveyor belts, air separation)
 - How will you ensure your device is efficient, safe, and scalable?
4. **Testing & Improvement:**
 - How will you measure the success of your design? (Accuracy, speed, cost)
 - What modifications can you make to improve performance?

Social-Emotional Learning (SEL) Relations:

1. **Collaboration & Teamwork** – Students work in groups, assigning roles (designer, builder, tester) and practicing active listening and compromise.
2. **Resilience & Growth Mindset** – Emphasizing that failure is part of the EDP; students iterate and improve without frustration.
3. **Responsible Decision-Making** – Discussing the environmental impact of recycling and ethical design choices (sustainability, cost, accessibility).
4. **Communication Skills** – Presenting designs, giving constructive feedback, and justifying engineering choices.

Lesson Plan

1. Engage

teacher Role:

1. Introduce the Problem:

- Present the challenge: *"We need to design a way to separate metals from mixed materials efficiently."*
- Use a **short video** (e.g., recycling plant struggles, e-waste pollution) or **images** of landfills with mixed metals.
- We can use this video about the difference between magnetic and non-magnetic materials. [<https://youtu.be/yorUUI5oCIE?si=H-qytdhQUBjWp82D>]

2. Ask Engaging Questions:

- *"Why is it hard to sort metals manually?"*
- *"What happens if metals aren't recycled properly?"* (Pollution, resource waste.)
- *"How might engineers solve this problem?"*

3. Activate Prior Knowledge:

- Ask: *"How can you tell if something is metal? What properties do metals share?"*
- Encourage students to brainstorm examples (magnets attracting some metals, conductivity, shine).

4. Show Empathy:

- *"Imagine if toxic metals from old phones ended up in water supplies—how would that affect people or animals?"*
- Connect to real-world impacts (e.g., mining harms ecosystems; recycling reduces waste).

5. Provide Context:

- *"Material scientists and engineers design solutions like this every day. Today, YOU will think like them!"*

- Briefly explain roles: *"Recycling engineers use magnets, sensors, and machines—but we'll start simple."*

Expected from Students:

1. Active Participation:

- Respond to questions (e.g., *"Why can't we use magnets for all metals?"*).
- Vote on ideas (e.g., *"Which sorting method seems most practical: magnets, density, or color?"*).

2. Share Prior Knowledge:

- *"Metals are shiny and hard!"* → Guide them to refine (e.g., *"But some plastics look metallic—how can we tell the difference?"*).

3. Curiosity and Inquiry:

- Ask questions like: *"Could we use water to separate metals?"* or *"Why does aluminum foil stick to magnets sometimes?"*

4. Empathy:

- Discuss: *"How does improper recycling affect your community?"*
- *"If you were a recycling worker, what would frustrate you about sorting?"*

5. Interest in the Task:

- Show excitement: *"Your design could help reduce pollution—let's prototype!"*
- Students might say: *"Can we build a conveyor belt like in the video?"*

Engineering design Process

Quickly review the phases of the Engineering Design Process, then encourage students to focus on the Ask and Brainstorming stages until they settle on a project idea. Remind them that there are no right or wrong answers for this step—it's all about exploring possibilities!

Engineering Design Process (EDP) for a Material-Sorting Machine

Problem Statement:

"Design a machine or device that can automatically sort different materials (e.g., metals, plastics, paper) for recycling or waste management."

EDP Step-by-Step Guide with Possible Answers

1. Ask (Define the Problem)

Teacher Prompts:

- *"Why is sorting materials important?"*
- *"What challenges do recycling centers face when separating materials?"*
- *"What materials are hardest to sort? Why?"*

Possible Student Answers:

- *"Sorting helps recycle materials properly."*
- *"Manual sorting is slow and expensive."*
- *"Some materials look similar but are different (e.g., aluminum vs. shiny plastic)."*

2. Research (Learn About Materials & Existing Solutions)

Teacher Prompts:

- *"What properties can we use to separate materials?"*
- *"How do real recycling machines work?"*

Key Concepts & Answers:

Material Properties	Possible Sorting Method
Metals Magnetic, conductive, heavy	Magnets, eddy current separators
Plastics Light, float in water	Density separation (air/water streams)
Paper Light, absorbent	Air jets, optical sensors
Glass Heavy, brittle	Vibration sieves, crushing

Activity: Test materials with magnets, water (float/sink), or conductivity testers.

3. Imagine (Brainstorm Solutions)

Teacher Prompt:

"How could a machine detect and separate materials automatically?"

Possible Student Ideas:

- *"A conveyor belt with a magnet to pull out metals."*
- *"A fan that blows light materials (plastic/paper) into one bin."*
- *"A camera that detects colors/shapes and uses robot arms to sort."*
- *"A water tank where heavy materials sink and light ones float."*

4. Plan (Choose a Design & Sketch It)

Teacher Guidance:

- Have students draw their machine with labels.
- List materials needed (e.g., motors, sensors, magnets, cardboard for prototyping).

Example Design:

- **Magnetic + Air Sorter:**
 1. Conveyor belt moves mixed materials.
 2. Magnet pulls out ferrous metals (iron, steel).
 3. Air jet blows lightweight plastics/paper into a separate bin.
 4. Heavy non-magnetic metals (aluminum) fall into a third bin.

5. Create (Build a Prototype)

Materials Suggestions:

- Low-cost: Cardboard, LEGO, small DC motors, magnets, straws (for air jets).
- Advanced: Arduino with color/conductivity sensors (for automated sorting).

Teacher Tips:

- Start simple (e.g., hand-cranked conveyor + magnet).
- Test after each modification!

6. Test & Improve (Iterate for Better Results)

Teacher Prompts:

- "Did your machine sort all materials correctly?"
- "What didn't work? How can you fix it?"

Example Improvements:

- "Our magnet missed small aluminum cans → Add an eddy current separator."
- "Plastic and paper stuck together → Increase air pressure or add a vibrating screen."

Real-World Connections

- Show videos of industrial sorting machines (optical sorters, magnetic separators).
- Discuss careers: "Recycling engineers design these systems—could that be you?"

Assessment Ideas

- **Success Criteria:** Does the machine correctly sort 3+ materials?
- **Presentation:** Students explain their design choices and improvements.

Practice



Decide which materials are best for different parts of the sorting machine.

Part	Best material
The base 	
The pulley 	
Belt 	
The body 	

Theme 1

[Sorting magnetic and non-magnetic materials]

UNIT 01:

Chapter 01: [Mechanics]

☐ Lesson 04: pulleys

☐ Time frame: 80 mins.

Objectives of the lesson:

1. Identify what a pulley is (a simple machine with a wheel and rope).
2. Describe how pulleys help lift or move objects with less effort.
3. Give examples of where pulleys are used in everyday life (e.g., flagpoles, cranes, blinds).
4. Demonstrate how a simple pulley works using hands-on materials (if possible).

Challenge Questions About Pulleys:

1. Basic Understanding:

- o How does a pulley make it easier to lift a heavy toy?
- o Can you name two places where we see pulleys in real life?

2. Design Thinking:

- o If you wanted to lift a bucket of blocks up high, how could a pulley help?
- o What happens if we use two pulleys instead of one?

3. Sorting Machine Challenge:

- o How could we use a pulley to move objects from a low table to a higher bin?
- o Can you design a pulley system that sorts big objects (like balls) from small ones (like marbles)?

- o What materials could we use to build a simple pulley sorting machine? (e.g., string, cups, cardboard tubes, toy wheels)

SEL relations:

1. Self-Awareness

- Example: Students reflect on their own ideas ("How can I design a pulley to sort objects?") and recognize their strengths/challenges during the activity.
- Prompt: "What part of building the pulley was easy/hard for you?"

2. Self-Management

- Example: Students practice patience and persistence when testing their pulley systems (e.g., troubleshooting if the rope slips).
- Prompt: "How did you stay calm when your first design didn't work?"

3. Social Awareness

- Example: Students learn about real-world uses of pulleys (e.g., construction cranes, wells in some communities) and discuss how they help people.
- Prompt: "How might a pulley help someone who can't lift heavy things?"

4. Relationship Skills

- Example: Working in pairs/groups to build the sorting machine, sharing materials, and giving feedback.
- Prompt: "How did you and your partner agree on a design? What did you do if you disagreed?"

5. Responsible Decision-Making

- Example: Choosing materials wisely (e.g., "Should we use a string or a rubber band for the pulley?") and testing ideas safely.
- Prompt: "Why did your group decide to change your design? How did it help?"

Lesson plan:

Engage:

Teacher Role:

1. Introduce the Problem:

- o Show a video of a recycling plant struggling to sort heavy/mixed materials (e.g., this pulley video).
- o Challenge: "How can we move heavy objects to sort them without lifting them by hand?"

2. Ask Engaging Questions:

- o "Why is it hard to lift a bucket of rocks? What if we could pull it up another way?"
- o "Where have you seen pulleys before? (Wells? Cranes?)"

3. Activate Prior Knowledge:

- o Have students try lifting a heavy book with/without a string looped over a chair (simulating a pulley).

- o "Which way felt easier? Why?"

4. Show Empathy:

- o "Imagine you're a worker sorting recycling—how could a pulley help your back not hurt?"

5. Provide Context:

- o "Today, you'll be engineers designing a pulley machine to sort objects!"

Student Role:

- Participate in discussions (e.g., shout out places they've seen pulleys).
- Share experiences: "My dad uses a pulley to lift our camping gear!"
- Ask questions: "Can pulleys push things too?"

This website has a game about simple machine:
<https://edheads.org/play-simple-machines/>

For this part we can use this video : <https://youtu.be/WSY4HzWZllo?si=X52dIrx6bqRdl-f7>

This video shows the steps how to make it :
<https://www.youtube.com/watch?v=1nBzmYxD-kI>
the steps of sorting machine:
https://docs.google.com/document/d/1-mnNiK99Ttt4rZXTQVVHtB_4qT23uYb5/edit?tab=t.0

Theme 1

[Sorting magnetic and non-magnetic materials]

UNIT 01:

Chapter 02: [Design]

- **Lesson 01: Replace Old pulley**
- **Time frame: 80 mins.**

Lesson Objectives:

By the end of this lesson, students will be able to:

1. **Distinguish** between 2D (flat) and 3D (solid) shapes by sorting examples (e.g., circle vs. sphere, square vs. cube).
2. **Create** a simple 3D shape (like a Keychain) using Tinkercad's basic tools.
3. **Define** "dimensions" (length, width, height) and adjust them in Tinkercad to modify shapes.
4. **Rotate** 3D shapes in Tinkercad to visualize different perspectives (e.g., flipping a cone upside down).
5. **Design** a simple pulley system in Tinkercad by combining shapes (e.g., wheel, axle, rope groove).

Challenges questions:

- *"Is a pizza slice a 2D or 3D shape? What about the whole pizza box?"*
- *"Why can't a 2D shape hold objects like a 3D shape can?"*
- *"How would you turn a 2D square into a 3D cube in Tinkercad?"*
- *"Why might you need to adjust the 'hole' tool when designing a pulley?"*

- *"Which dimension (length/width/height) would you change to make a tall, skinny tower?"*
- *"Why is it important to check all sides of your shape (front/back/top) before printing?"*
- *"What's the simplest shape you could use for a pulley wheel? Why?"*
- *"Where should you place the 'axle' to make your pulley spin smoothly?"*
- *"What would happen if your pulley wheel had no groove for the rope?"*

Reflection Questions (SEL Connection):

- *"What was the hardest part of designing in Tinkercad? How did you solve it?"*
- *"If you could build another simple machine (like a lever), which would you choose and why?"*

Lesson plan :

Engage :

Teacher Role:

1. Spark Curiosity:

- Show real-world examples (e.g., hold up a paper circle vs. a ball, or play [this](https://www.youtube.com/watch?v=Acr2bu9FbPM) <https://www.youtube.com/watch?v=Acr2bu9FbPM> video).
- They can play this game: <https://wordwall.net/resource/13362759/math/2d-vs-3d-shapes>
- Ask: *"Why can't a 2D shape roll like a 3D one?"*

2. Introduce the Challenge:

- *"Today, we'll design a 3D pulley in Tinkercad to lift heavy objects—but first, let's explore shapes!"*
- Show a simple pulley animation (e.g., [this one](#)).

3. Ask Guiding Questions:

- *"How could a 3D wheel solve a problem a 2D circle can't?"*
- *"What parts would a pulley need to spin smoothly?"*

4. Activate Prior Knowledge:

- Have students sort objects into 2D (paper cutouts) and 3D (blocks, cans) bins.
- *"How are the shapes in your pencil case (like erasers) different from drawings in your notebook?"*

5. Set Up the Tech:

- Demo Tinkercad's interface briefly (*"This is where we'll build our pulley!"*).

Student Role:

let's
Think



How do you think we can make
this pulley using Tinkercad?
What basic shapes can we use?



1. Observe and Respond:

- Answer questions (e.g., "A 2D shape is flat; 3D shapes are thick!").
- Share experiences: "I saw a pulley at the flagpole!"

2. Explore Hands-On:

- Manipulate 2D/3D objects (e.g., roll a square).
- Predict: "If I turn this 2D net into a cube, marbles?"



playground on the

sphere, trace a

will it hold

3. Think Critically:

- Compare shapes: "Why does a 3D cylinder work better as a pulley wheel than a flat circle?"
- Brainstorm: "What else could we lift with a Tinkercad pulley?"

4. Ask Questions:

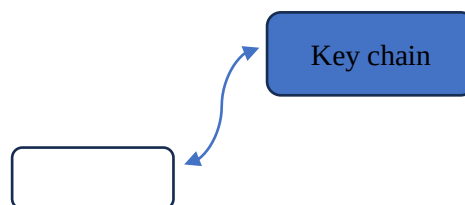
- "Can we make a pulley with triangles?"
- "How do we add a 'rope' to our design?"

Steps to make the sandwich keychain:

• Materials

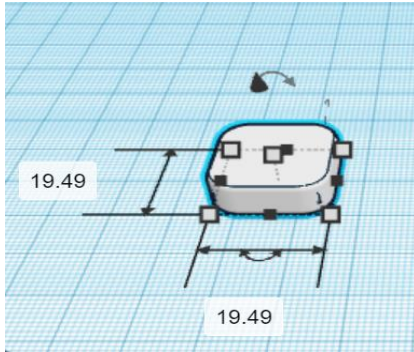
1. Laptops

2. TINKERCAD class code

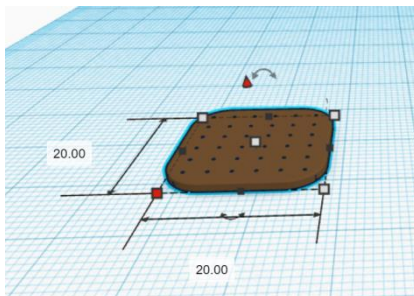


- **STEPS**

1. Drag a square with dimensions L: 20 W: 20 H: 1.5

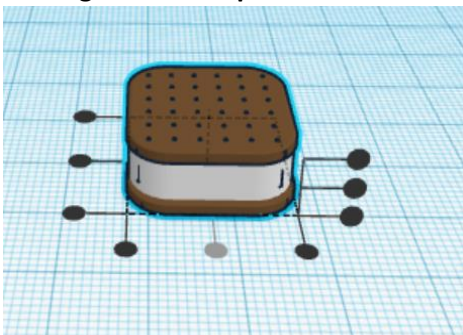


2. Drag another square with dimensions L: 19.49 W: 19.49 H: 6.20

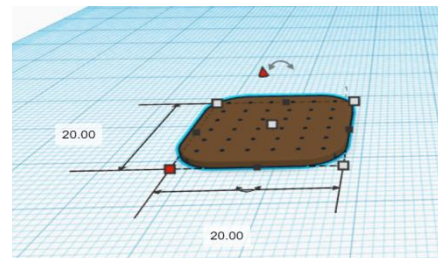


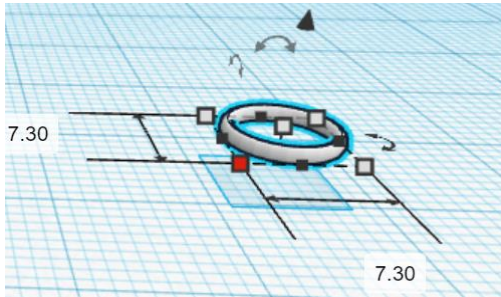
3. Take a copy from the first square but raise it by 6

4. Let's align our three parts

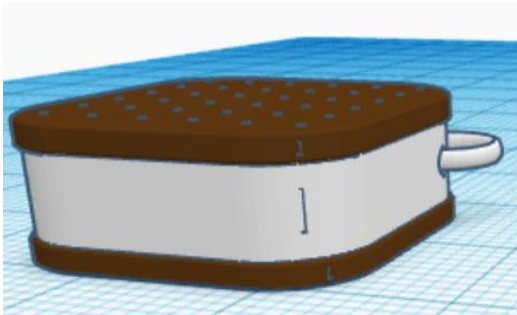


together





5. Drag a ring and make its dimensions L:7.30 W:7.30 H:1 and raise it by 3



6. Place the ring to your sandwich

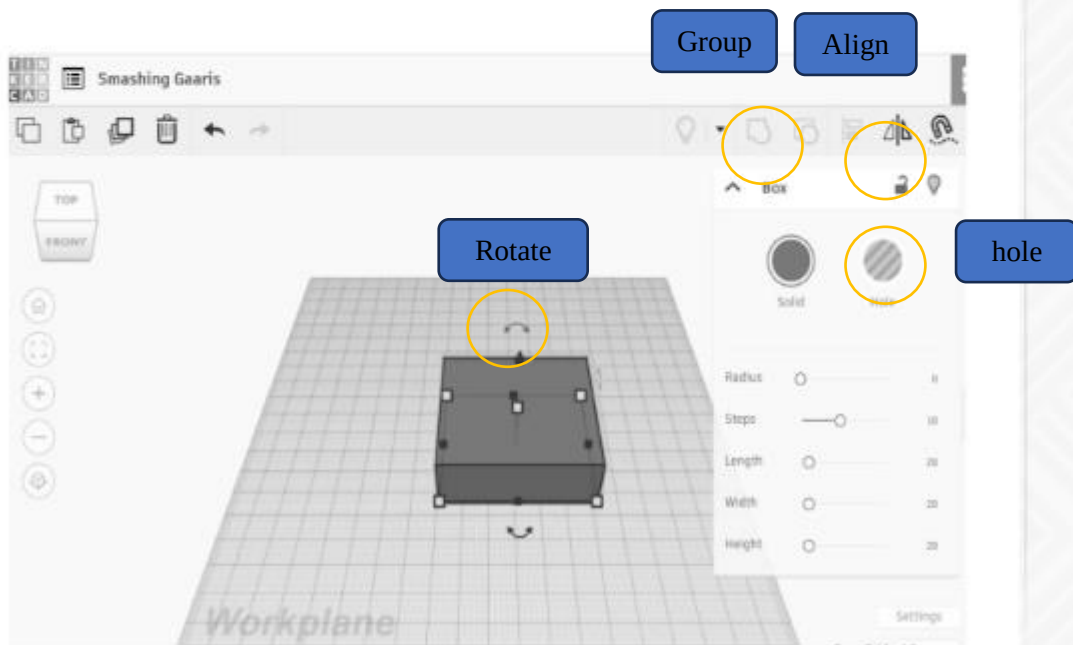
Assessment

Focus



Draw a circle around :

1. The tool we use to rotate shapes.
2. Tool to group shapes.
3. Tool to make holes.
4. Tool to align things.



We can use this game to know the different features of Tinker cad :

<https://wordwall.net/resource/56796670/tinkercad-features> \

Practice



Design your pulley using tinkercad.



Steps to make the pulley:

<https://docs.google.com/document/d/1-PWABRrAiWI-mdflvyEg9LsWN8Lkxe80/edit>

Theme 1

[Sorting magnetic and non-magnetic materials]

UNIT 01:

Chapter 02: [Design]

- Lesson 02: Electricity Domain
- Time frame: 80 mins.

Learning Objectives:

- ✓ Define electricity, circuits, and open/closed Circuits with real-world examples.
- ✓ Build a functional circuit with a buzzer + switch and troubleshoot failures.
- ✓ Explain how DC motors work and integrate one into their sorting machine.
- ✓ Collaborate to refine designs (e.g., adjust motor placement for efficiency).

challenges questions:

Basic Circuit Challenges (Recall & Troubleshooting)

1. "Your buzzer isn't working. List 3 possible reasons why."
(Expected: Dead battery, loose wire, broken buzzer, switch not pressed, etc.)
 2. "Draw an open circuit and a closed circuit. Which one allows electricity to flow?"
(Assesses understanding of circuit completion.)
 3. "If you remove the push button from your circuit, will the buzzer turn on? Why or why not?"
(Tests grasp of switch function—without it, circuit is either always on or open.)
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Motor & Design Challenges (Application)

4. "Your DC motor spins too slowly. How could you make it faster without adding more batteries?"
(Possible solutions: Reduce friction, use a gear/pulley, check for wire resistance.)
 5. "How would you modify your circuit to make the motor start ONLY when a button is pressed AND a light is on?"
(Tests understanding of series circuits—students must connect button + LED + motor in series.)
 6. "Your motor works, but your sorting machine jams when items pile up. How could you fix this?"
(Encourages mechanical problem-solving: e.g., widen the chute, add a funnel, slow the motor.)
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Advanced Challenges (Critical Thinking & Creativity)

7. "Can you design a circuit where one button turns the motor ON and another turns it OFF?"
(Hint: Requires two switches in parallel—advanced students might mimic a latching relay.)
8. "How could you use a motor to sort objects by WEIGHT (not just size/shape)?"
(Creative solutions: Motor tilts a platform, spins a wheel with gaps, etc.)
9. "What real-world device uses a circuit like yours? How is it similar or different?"
(Connects to engineering—e.g., elevator buttons, car windows, conveyor belts.)
10. "If your motor had to run on solar power, what changes would you make to your circuit?"
(Explores energy sources—may need capacitors, larger solar panels, or energy-efficient motors.)

● SEL relations:

SOCIAL (Collaboration & Community)

- **Structured Group Roles:** Assign rotating tasks (e.g., Wire Technician, Battery Checker, Motor Director) to ensure equitable participation.
 - **Gallery Walk:** Groups showcase motorized sorting machines while peers give "Kind-Question-Suggestion" feedback (e.g., "How did you attach the motor?").
 - **Real-World Links:** Discuss how electricians/engineers collaborate (e.g., teams building robots).
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EMOTIONAL (Mindset & Wellbeing)

- **"Fail Forward" Ritual:** Start class with a 2-minute "My Best Mistake" share (e.g., "Yesterday, I connected wires backward—now I know polarity matters!").

- **Anxiety Anchors:** Provide "*Circuit Safety Nets*"—pre-tested spare parts for frustrated students to swap in.
- **Progress Tracking:** Use emoji scales (🤔→🚀) for students to self-rate confidence before/after activities.

LANGUAGE (Literacy & Communication)

- **Tiered Vocabulary:**
 - **Basic:** *Circuit, battery, spin*
 - **Advanced:** *Polarity, conductivity, mechanical energy*
- **Dialogic Prompts:**
 - *"Explain to your partner how the buzzer circuit works using the words flow and break."*
 - *"Compare your motor setup to a household device (e.g., blender, fan)."*
- **Multilingual Supports:** Use GIFs (e.g., looping wire connections) + Google Translate for key terms.

Lesson plan:

Engage:



Teacher Role:

1. Prompt Discussion (5 mins):

- **Ask:** *"What is electricity? Where do we see it in daily life?"*
- **Write student ideas on the board** (e.g., lights, phones, batteries).
- **Define electricity simply:** *"The flow of tiny particles called electrons that power devices."*

2. Introduce Circuits (5 mins):

- **Show a battery, wire, and buzzer:** *"How could we make this buzzer work?"*
- **Explain:** *"A circuit is a closed loop that electricity flows through. If it's broken (open circuit), nothing happens!"*

3. Video Hook (5 mins):

- **Play** <https://youtu.be/HOFp8bHTN30?si=feOFSJxxYcREwKFY> (0:00–2:30).

Pause to highlight:

- *"See how the switch controls the flow?"*
- *"What happens when the circuit is incomplete?"*

4. Game-Based Practice (10 mins):

- Guide students to play [Crack the Circuit](#) (Level 1).
- Walk around, ask: *"Why did your solution work? What's the path of the current?"*

Student Role:

1. Brainstorm & Share:

- Respond to the teacher's questions (e.g., *"Electricity powers my Xbox!"*).
- Predict: *"If I connect the battery to the buzzer, will it make sound? Why?"*

2. Watch & Analyze:

- Note how the video shows open/closed circuits.
- Turn-and-talk: *"What's one thing you learned from the video?"*

3. Play & Apply:

- Test virtual circuits in the game.
- Compare strategies with peers: *"I connected the battery first—what did you do?"*


**let's
Think**



How do you think toys and fans rotate?

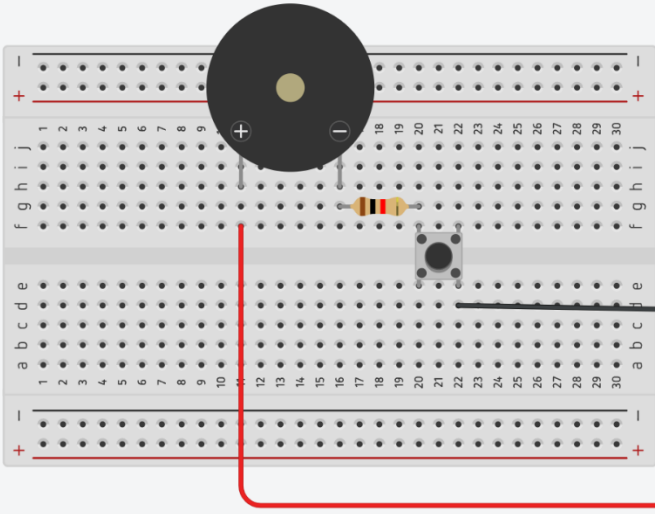
1. "Toys work with batteries that give power to a tiny motor inside!"
2. "Fans use electricity to spin their motor and blades, making wind."
3. "Motors make things move when electricity flows through them!"



Let's make the first circuit:
Needed Materials:

- Batteries
- wires
- Buzzer
- push button
- resistor.

Connection on Tinker-cad:



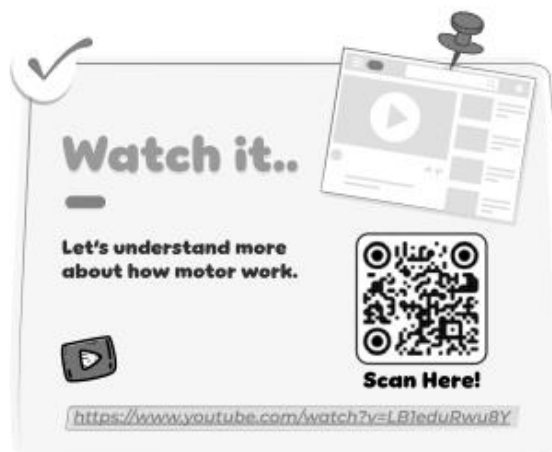
Explore



Do the activity and close the circuit. What do you see? Why do you think this is happening?

.....

.....



In this part we can use this Video:

➔ https://youtu.be/LB1eduRwu8Y?si=waFZ9e_Y_6uy6eW

for the activity in this file (fifth activity)

<https://docs.google.com/document/d/1-PWABRRrAiWl-mdflvyEg9LsWN8Lkxe80/edit>

ASSESSMENT

Focus



Choose the correct answer from the choices for each question.

1. What do motors change electricity into?

- A) Light.
- B) Sound.
- C) Movement.
- D) Heat.

2. Which of these likely does NOT use a motor?

- A) A toy car.
- B) A pencil.
- C) A washing machine.
- D) An electric toothbrush.

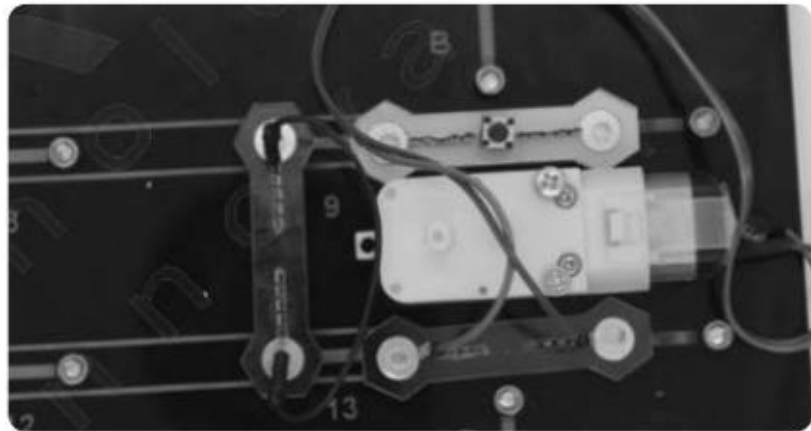
3. What is one thing that motors help do?

- A) Make things stay still.
- B) Make things move.
- C) Make things disappear.
- D) Make things bigger.

Practice



Create a motor circuit on smartpad.



**After this practice attach the motor to the sorting machine.
for the Circuit in this file (fifth activity)**

<https://docs.google.com/document/d/1-PWABRRrAiWl-mdflvyEg9LsWN8Lkxe80/edit>